# Ramp energy lab



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Compare gravitational energy change with measured kinetic energy.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 4, Day 10; 3 course days later.*
K=12 J.

**02. Explain a worked example (worked)**
U loss 6=translational gain 4+other 2; do not automatically call every difference measurement error.

**03. Look and explain (model)**
Vertical height.

**04. Build the solution (practice)**
Lost U 5 J; difference 1 J.

**05. Use the idea (reason)**
Teacher checks measured data, units and clear reference; no invented results.

**06. Choose a method (mixed)** *Review: Unit 4, Day 1; 12 course days later.*
Factor 16, because $K=\frac{1}{2}mv^2$.

**07. Exit check (exit)**
Rotational energy and thermal energy; others justified.

**08. STRETCH: Apply it another way (stretch)**
Measure rotation or compare a controlled sliding setup, keep heights/masses consistent and track uncertainties.

**Teaching note:** Investigation companion. Supplied values are for analysis only; record actual results separately.

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**Learning target:** Compare gravitational energy change with measured kinetic energy.

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**01. Bring back an earlier idea (retrieve)** *Review: Unit 4, Day 10; 3 course days later.*
$h=v_i^2/(2\ g)=11.25$ m.

**02. Explain a worked example (worked)**
U loss 6=translational gain 4+other 2; do not automatically call every difference measurement error.

**03. Represent and interpret (model)**
Vertical height.

**04. Solve and justify (practice)**
Initial U=.98 J; translational K 1.5625 J, too large for assumptions. Check push, height, mass, speed calibration/location and uncertainty.

**05. Analyze the reasoning (reason)**
Teacher checks measured data, units and clear reference; no invented results.

**06. Choose a method (mixed)** *Review: Unit 4, Day 1; 12 course days later.*
K becomes 1/4 of the original; fixed p means $K \propto 1/m$.

**07. Exit check (exit)**
1.96 J; 1.44 J; 73.5%.

**08. STRETCH: Apply it another way (stretch)**
U drop 2 J; K 1 J; remainder 1 J. Conditional friction magnitude 1 N; rotation would alter that attribution.

**Teaching note:** Investigation companion. Supplied values are for analysis only; record actual results separately.